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| **Course** | MCA -1 |
| **Subject Name** | OPERATING SYSTEMS |
| **Subject Code** | 05MC1104 |
| **Submission Date** | 30-10-26 |

## LAB Manual Questions

| Sr. No. | List of Practical / Exercise | CO |
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| 1 | Introduction to Linux Environment: Execute and analyze basic Linux commands such as date, ls, pwd, who, cal, ps, uname, mkdir, rmdir, cd, cp, mv, rm, cat, wc, and chmod. | CO1 |
| 2 | Write a shell script to accept a number and determine whether it is even or odd, calculate the sum of its digits, and find the number of digits. | CO1 |
| 3 | Write a shell script to accept a string and perform operations such as checking whether it is a palindrome, counting characters, and converting it between uppercase and lowercase. | CO1 |
| 4 | Write a shell script to search for a specified file in a directory and its subdirectories and display an appropriate message depending on whether the file is found or not. | CO2 |
| 5 | Write a shell script to display the currently running processes at regular intervals using both while and for loops. | CO2 |
| 6 | Write a shell script to read student roll number, name, and marks of three subjects from a file and generate a formatted marksheet containing total, percentage, and result. | CO2 |
| 7 | Write a shell script to accept a filename and display its permissions, ownership, and file type. Display an appropriate message if the file does not exist. | CO3 |
| 8 | Develop a menu-driven shell script to perform file and directory operations such as displaying the current directory, listing files, creating and changing directories, copying, renaming, deleting, and displaying files. | CO3 |
| 9 | Write a shell script to display a list of all files in the current directory for which you have read, write and execute permissions | CO3 |
| 10 | Write a script to display the date, time and a welcome message (Good Morning/Afternoon/Evening) using 24-hour notation | CO3 |
| 11 | Write a Shell Script to take user data as command line argument and display a greetings message | CO4 |

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| 12 | Write a shell script to accept distance between two cities (in km) and convert it to meters, feet, inches and centimeters | CO4 |
| 13 | Write a shell script to print all the prime numbers from 1 to 300 (Use nested loops, break and continue) | CO5 |
| 14 | Write shell script to accept the string and replace it by another string | CO5 |
| 15 | Write a script that deletes all leading and trailing spaces in all lines in a file. Also remove blank lines from the file | CO5 |

**Note:**

**1. All the students will have to submit the task before the given deadline.**

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